

WEEK7-SKILL

Task1:

To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs

```
UW PICO 5.09 File: input_redirect.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main()
{
    char input[200];
    char filename[100];

    printf("Enter input file name: ");
    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    strcpy(filename, input);

    int fd = open(filename, O_RDONLY);

    if (fd == -1)
    {
        perror("File");
        return 1;
    }

    pid_t pid = fork();

    if (pid == -1)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        dup2(fd, STDIN_FILENO);

        close(fd);

        execlp("cat", "cat", NULL);

        perror("cat");
        exit(1);
    }

    close(fd);

    waitpid(pid, NULL, 0);

    printf("Input redirection completed\n");

    return 0;
}
```

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```
vishalkarnati@VISHALS-MacBook-Air Desktop % cd os
vishalkarnati@VISHALS-MacBook-Air os % mkdir week7skill
vishalkarnati@VISHALS-MacBook-Air os % cd week7skill
vishalkarnati@VISHALS-MacBook-Air week7skill % nano input_redirect.c
vishalkarnati@VISHALS-MacBook-Air week7skill % gcc input_redirect.c -o input_redirect
vishalkarnati@VISHALS-MacBook-Air week7skill % ./input_redirect
Enter input file name: input.txt
file: No such file or directory
vishalkarnati@VISHALS-MacBook-Air week7skill % nano input.txt
vishalkarnati@VISHALS-MacBook-Air week7skill % ls
input_redirect      input_redirect.c    input.txt
vishalkarnati@VISHALS-MacBook-Air week7skill % ./input_redirect
Enter input file name: input.txt
Hello
This is an input redirection test.
Operating Systems
Input redirection completed
vishalkarnati@VISHALS-MacBook-Air week7skill %
```

Task2:

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection

```
UW PICO 5.09 File: output_redirect.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main()
{
    char filename[100];

    printf("Enter output file name: ");
    scanf("%s", filename);

    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("File");
        return 1;
    }

    pid_t pid = fork();

    if (pid == -1)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        dup2(fd, STDOUT_FILENO);
        close(fd);

        execlp("ls", "ls", NULL);

        perror("ls");
        exit(1);
    }

    close(fd);

    waitpid(pid, NULL, 0);

    printf("Output redirection completed\n");

    return 0;
}
```

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```
vishalkarnati@VISHALS-MacBook-Air week7skill % nano output_redirect.c
vishalkarnati@VISHALS-MacBook-Air week7skill % gcc output_redirect.c -o output_redirect
vishalkarnati@VISHALS-MacBook-Air week7skill % ./output_redirect
Enter output file name: output.txt
Output redirection completed
vishalkarnati@VISHALS-MacBook-Air week7skill % cat output.txt
input_redirect
input_redirect.c
input.txt
output_redirect
output_redirect.c
output.txt
vishalkarnati@VISHALS-MacBook-Air week7skill %
```

Task3:

To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios

```
UW PIC0 5.09 File: append_redirect.c Modified
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main()
{
    char filename[100];

    printf("Enter output file name: ");
    scanf("%s", filename);

    int fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);

    if (fd == -1)
    {
        perror("File");
        return 1;
    }

    pid_t pid = fork();

    if (pid == -1)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        dup2(fd, STDOUT_FILENO);

        close(fd);

        execlp("date", "date", NULL);

        perror("date");
        exit(1);
    }

    close(fd);

    waitpid(pid, NULL, 0);

    printf("Output appended successfully\n");

    return 0;
}
```

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```
vishalkarnati@VISHALS-MacBook-Air week7skill % nano append_redirect.c
vishalkarnati@VISHALS-MacBook-Air week7skill % gcc append_redirect.c -o append_redirect
vishalkarnati@VISHALS-MacBook-Air week7skill % ./append_redirect
Enter output file name: output.txt
Output appended successfully
vishalkarnati@VISHALS-MacBook-Air week7skill % cat output.txt
input_redirect
input_redirect.c
input.txt
output_redirect
output_redirect.c
output.txt
Thu Sep 17 22:52:32 IST 2026
vishalkarnati@VISHALS-MacBook-Air week7skill %
```

Task4:

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases

UW PICO 5.09File: stderr_redirect.cModified

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

int main()
{
    char filename[100];

    printf("Enter error output file name: ");
    scanf("%s", filename);

    int fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("File");
        return 1;
    }

    pid_t pid = fork();

    if (pid == -1)
    {
        perror("fork");
        close(fd);
        return 1;
    }

    if (pid == 0)
    {
        dup2(fd, STDERR_FILENO);
        close(fd);

        execlp("ls", "ls", "does_not_exist", NULL);

        perror("ls");
        exit(1);
    }

    close(fd);

    waitpid(pid, NULL, 0);

    printf("Error redirection completed\n");

    return 0;
}
```

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```
vishalkarnati@VISHALS-MacBook-Air week7skill % nano stderr_redirect.c
vishalkarnati@VISHALS-MacBook-Air week7skill % gcc stderr_redirect.c -o stderr_redirect
vishalkarnati@VISHALS-MacBook-Air week7skill % ./stderr_redirect
Enter error output file name: error.txt
Error redirection completed
vishalkarnati@VISHALS-MacBook-Air week7skill % cat error.txt
ls: does_not_exist: No such file or directory
vishalkarnati@VISHALS-MacBook-Air week7skill %
```

